

E-COMMERCE DYNAMIC PRICING AND ELASTICITY ANALYTICS

Overall Project Documentation

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Project Type: Data Analytics Project

Tools: Python, Excel, SQL, Power BI

Dataset: Apple Products Pricing (2020–2026)

Project Overview

This project focuses on analyzing e-commerce product pricing data to understand pricing patterns, discounts, sale events, product categories, stock status, ratings, and customer reviews. The analysis uses Python, Excel, SQL, and Power BI to transform raw data into meaningful business insights and support data-driven pricing and inventory decisions.

Project Objectives

- Analyze e-commerce product pricing patterns.
- Compare launch prices with current prices.
- Analyze discounts and sale events.
- Study product categories and stock status.
- Analyze ratings and reviews.
- Perform data analysis using Python and SQL.
- Develop an interactive Power BI dashboard.
- Generate business insights and recommendations.

Dataset Description

The project uses the Apple Products Pricing dataset covering the period 2020–2026. The dataset contains approximately 80,000 records and 14 columns. It includes product, pricing, discount, sale-event, stock, rating, and review information

Number of records: ~80,000

Number of columns: 14

A separate Data Dictionary is included in the project folder to describe each column in detail.

Excel Analysis

Excel was used for initial data exploration and analysis. Pivot Tables were created to summarize and compare the dataset across different categories and attributes. This helped in understanding the data structure and identifying useful patterns before performing detailed analysis using Python and SQL.

Tools & Technologies

Tool

Purpose

Excel	Pivot Tables and initial data analysis
Python	Data cleaning and EDA
Pandas & NumPy	Data manipulation and analysis
Matplotlib	Data visualization
MySQL	SQL querying and analysis
Power BI	Interactive dashboard development
GitHub	Project repository and version control

Project Methodology

Data Collection → Data Cleaning → Excel → EDA → SQL Analysis → Power BI

Dashboard → Insights → Recommendations

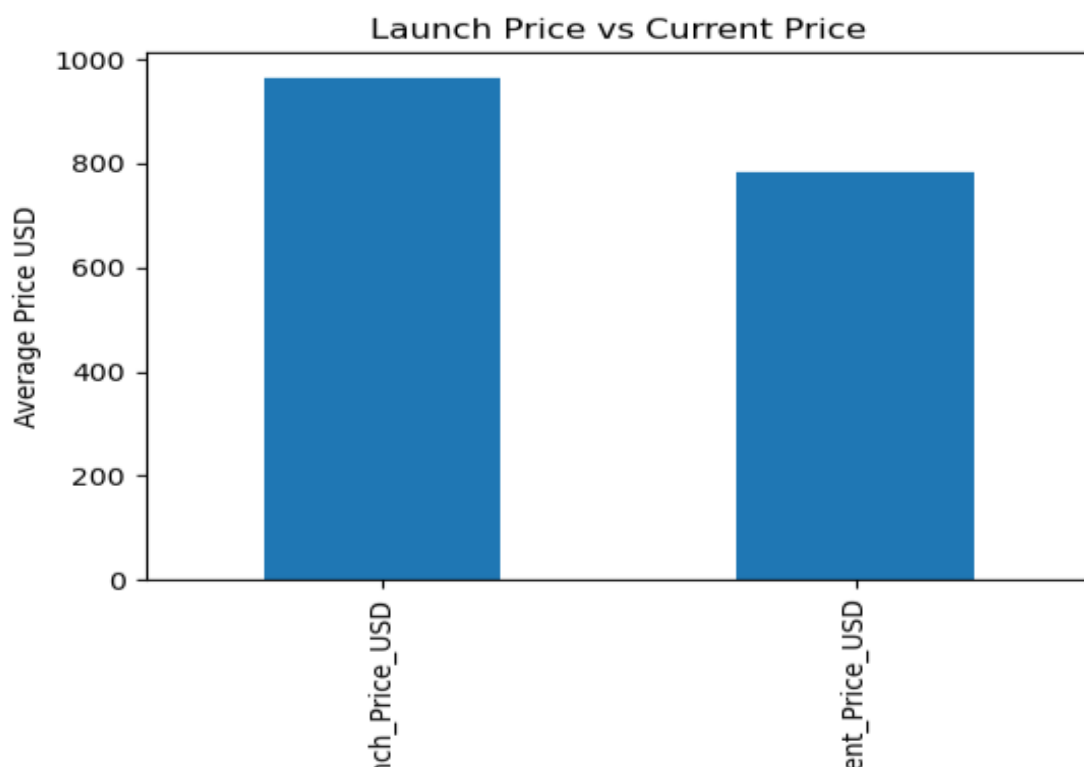
Data Cleaning and Preprocessing

- Checked missing values.
- Checked duplicate records.
- Corrected data types.
- Standardized data where required.
- Prepared the dataset for analysis and visualization.

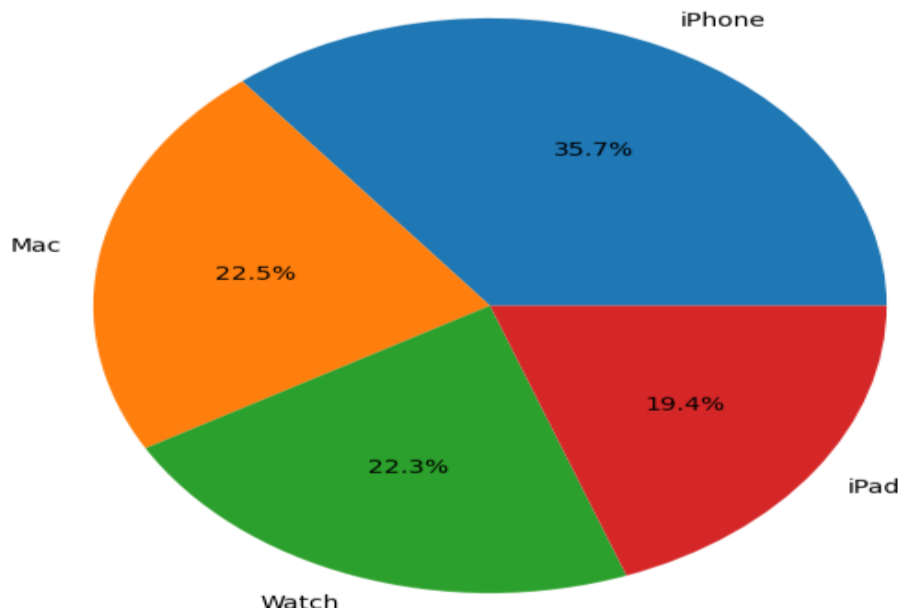
Exploratory Data Analysis(EDA)

Exploratory Data Analysis was performed to understand the distribution and relationships within the pricing data.

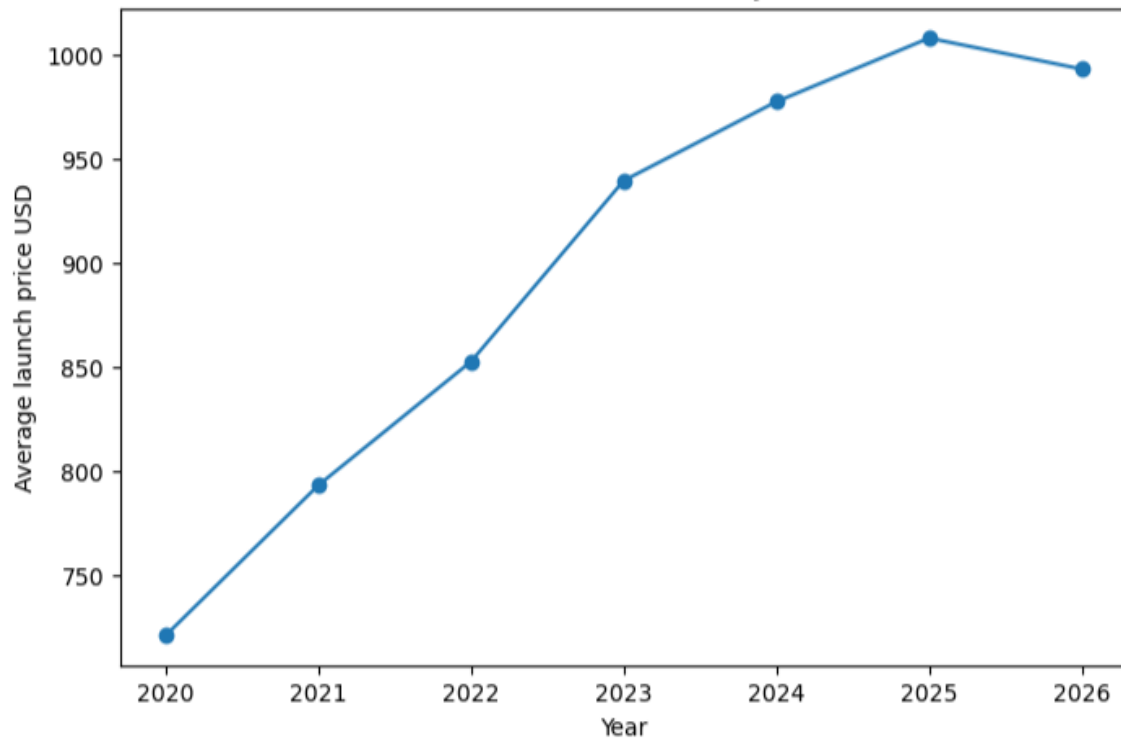
List of charts/analysis

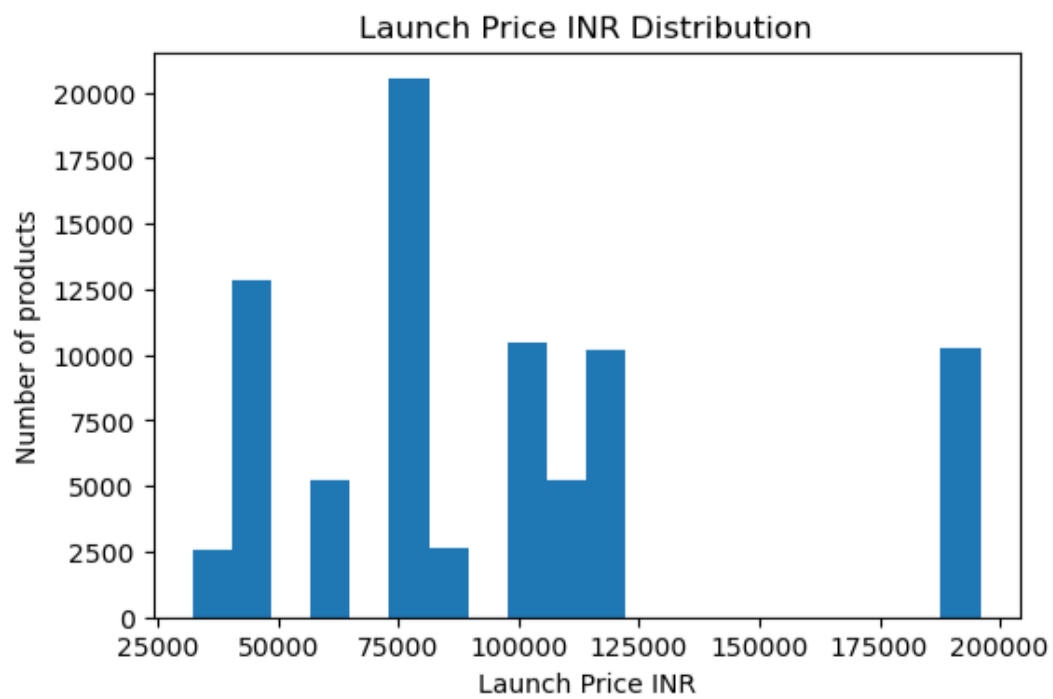
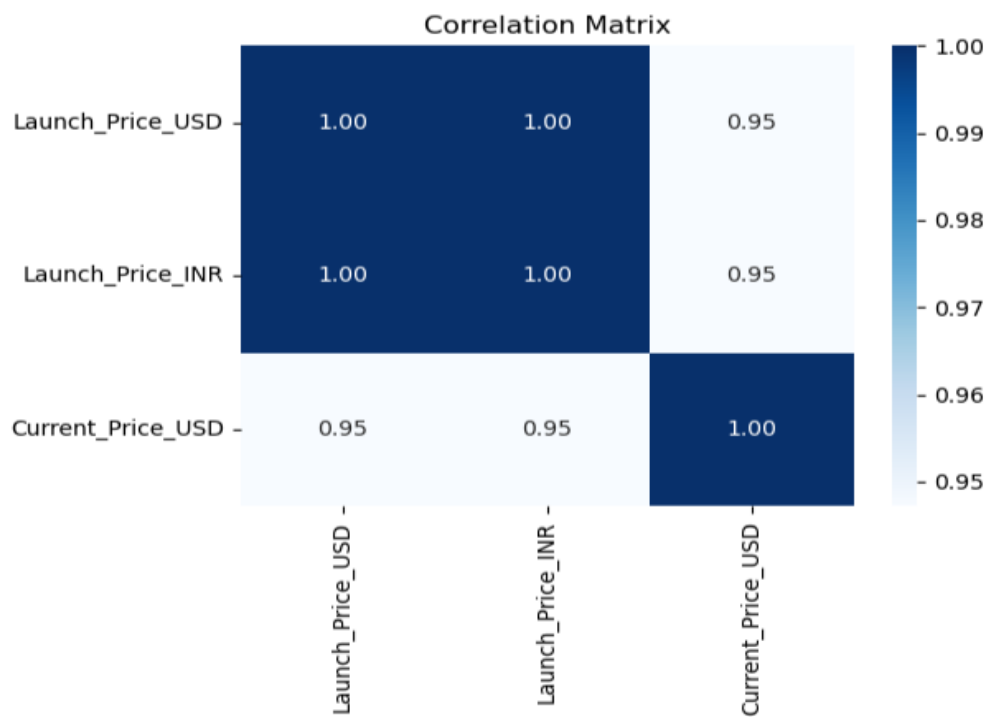


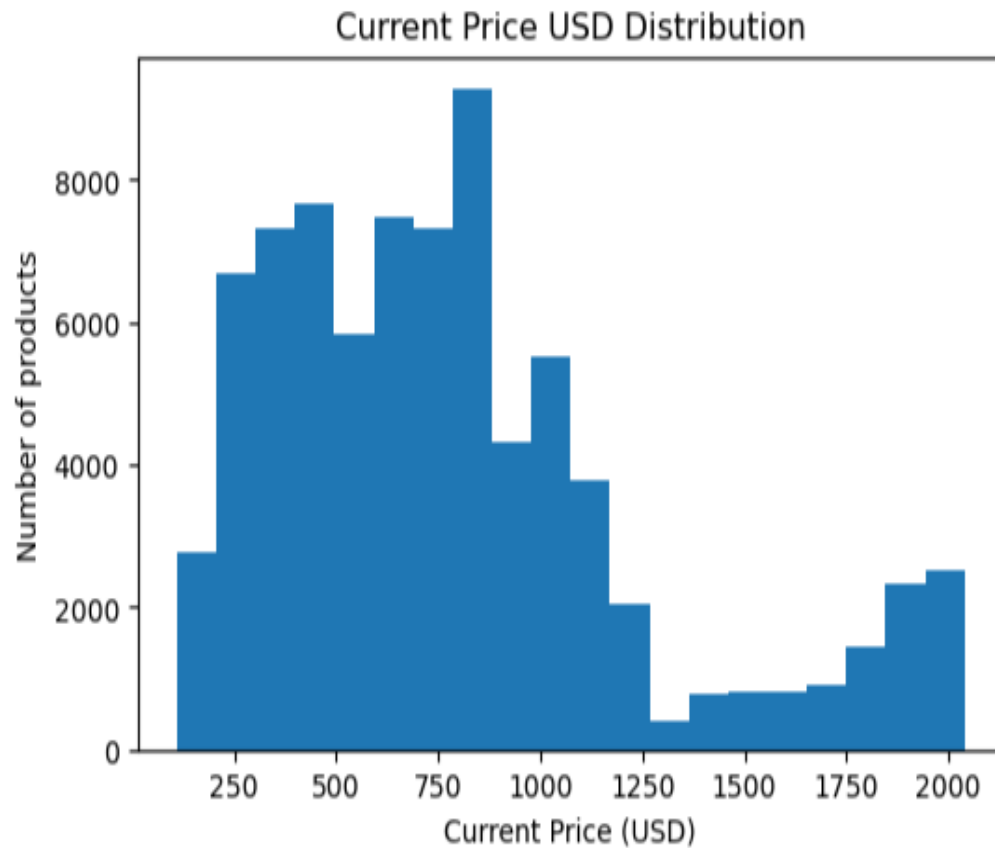
Product category composition



Launch Price Trend over years







SQL Analysis

SQL was used to perform structured analysis of the pricing dataset. Aggregate functions, GROUP BY, ORDER BY, joins, views, and window functions were used to extract useful information.

list:

COUNT() | SUM() | AVG() | MIN() | MAX()

RANK() | DENSE_RANK() | LAG() | LEAD()

Power BI Dashboard

Overview: KPI cards and overall pricing/product analysis.

Trends: Analysis of pricing trends over time.

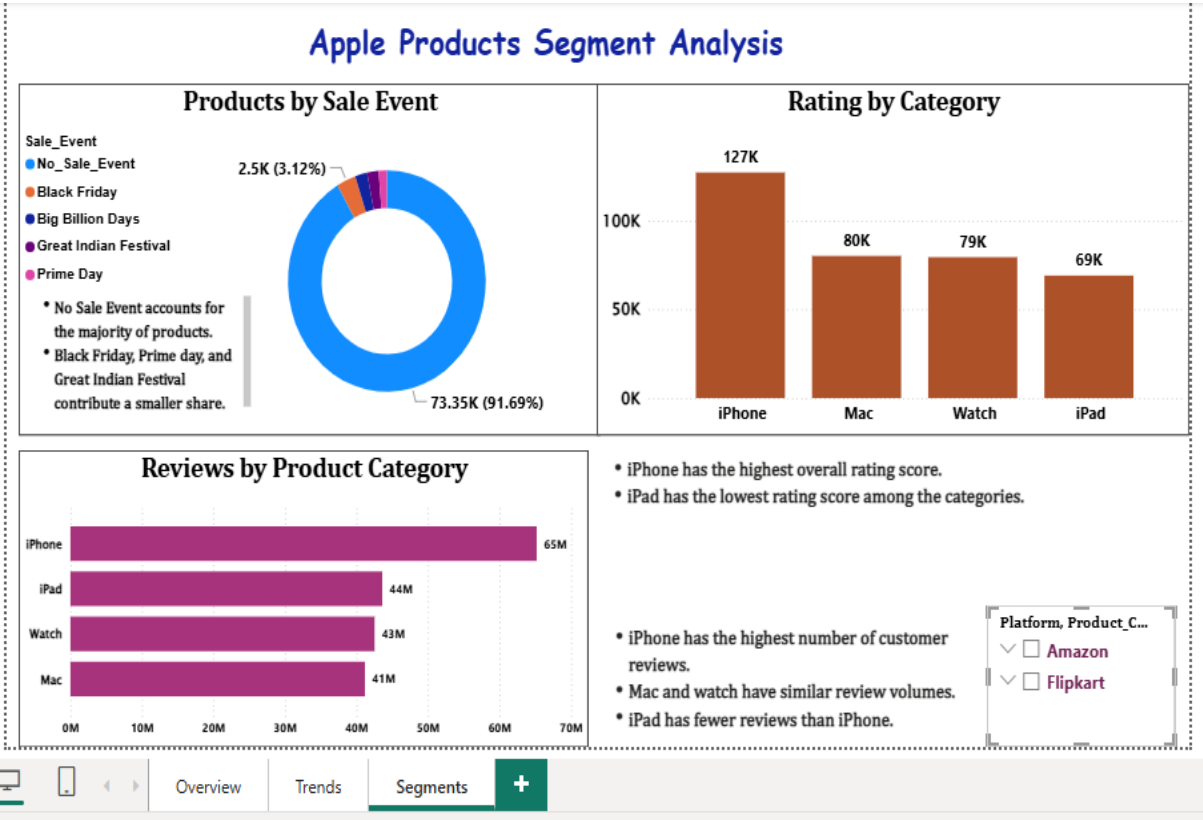
Segments: Product category and stock-status analysis.

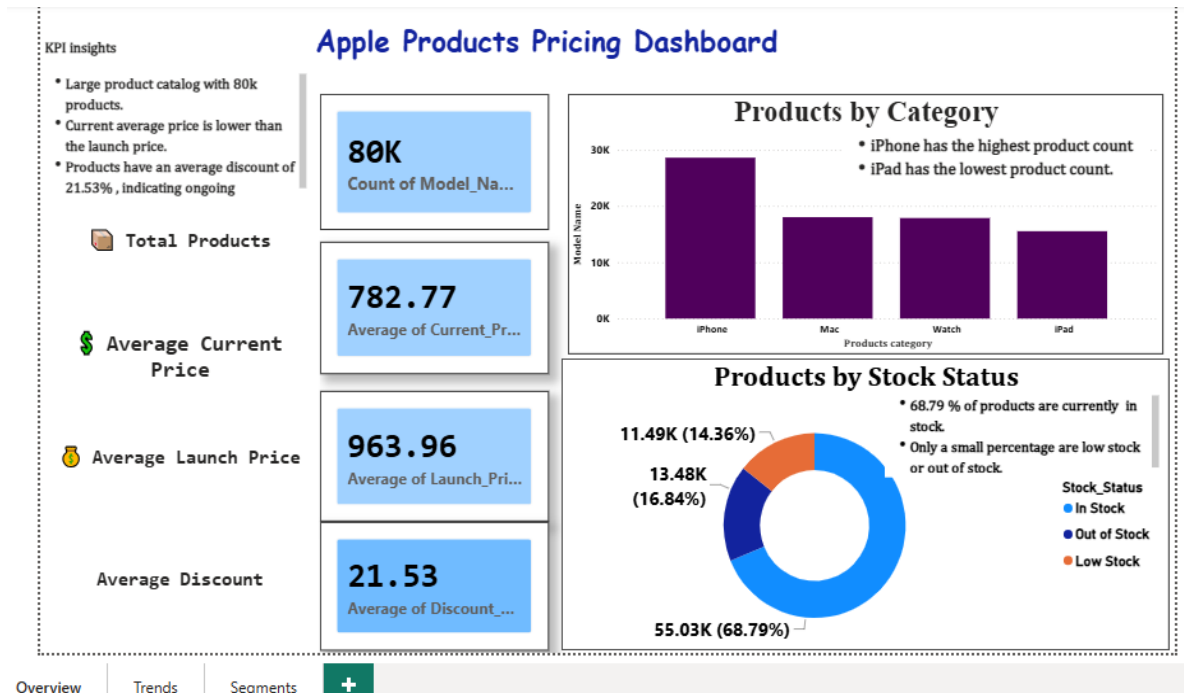
KPI cards

Charts

Slicers

Filters





Key Insights and Findings

Product prices vary significantly across categories and platforms.

Discounts and sale events influence current product prices.

Stock availability differs across product categories.

Customer ratings help identify better-performing product categories.

Interactive dashboards make pricing and product analysis easier for decision-making.

Business Recommendations

Use dynamic pricing strategies.

Optimize discounts based on pricing patterns.

Monitor inventory and stock status.

Apply category-based pricing strategies.

Consider customer ratings and reviews when making decisions.

Project Outcome

The project successfully transformed raw e-commerce pricing data into structured analysis and interactive visualizations. The combined use of Python, SQL, and Power BI helped identify pricing patterns and generate actionable business insights.

Conclusion

This project demonstrates how data analytics can be used to understand e-commerce

pricing behavior and support better business decisions. The analysis provides insights into pricing, discounts, product categories, sale events, and stock status through Python, SQL, and Power BI.

Supporting Documents

Data Dictionary

Data Cleaning Log

SQL Scripts

Python Notebooks

Power BI Dashboard

Dashboard Documentation

Project Presentation

README

GitHub Repository